

Research article

Application of a hybrid BWM-TOPSIS approach for mineral potential mapping

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ABSTRACT

Many mineral predictive models created based on multicriteria decision making (MCDM) methods use only one conceptually-based MCDM technique for data integration and synthesis of the mineral-related predictors. It is noteworthy that relying on just one mode of the conceptually-based data integration technique is often insufficient, as it fails to address the problems the other mode (in terms of either determining the weights of the predictors or by ranking and prioritising the predictors) deals with before the predictors are synthesised. Herein, a hybrid conceptually-based data integration approach comprising the best-worst method (BWM) and the Technique for Order of Preference by Similarity (TOPSIS) methods have been adopted in mapping viable regions of gold mineralisation occurrences over the Abansuoso Area of Ghana's Ashanti Region. The combined use of these two conceptually-based data integration approaches is rare in the literature, particularly in Ghana and West Africa. Based on the aforementioned approach, weights of nine predictors which were sourced from geological and geophysical datasets comprising magnetics and radiometrics were estimated using the BWM approach to determine their comparative importance towards the gold prospect of the study area. Afterwards, the TOPSIS approach was applied to prioritise and rank the various alternatives that makeup the predictors identified for this study. Subsequently, a predictive model that defines the spatial distribution of the mineral prospects of the study area was developed and was referred to as BWM-TOPSIS based mineral potential map (MPM). The BWM-TOPSIS-based MPM was further classified to characterise zones of low, moderate and high mineral potential, with their resulting areal extents being respectively 237.13 km², 225.66 km² and 127.45 km². An evaluation of the MPM developed based on the BWM-TOPSIS technique was conducted by implementing the prediction-area (P-A) plot. Outputs from the P-A plot indicate that 30 % of the study area has high prospect with 70 % of the existing gold locations observed within it. Additionally, the receiver operating characteristics (ROC) curve was used to evaluate the effectiveness of the MPM developed. The MPM generated based on the BWM-TOPSIS approach yielded an AUC score of 0.81; this AUC score indicates that the performance of the predictive model developed is very good.

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1. Introduction

Mineral resources in Ghana are an effective means of raising the country's income. Therefore, it is crucial to look for more ore resources. Ghana has attracted foreign investments for the exploration and exploitation of its mineral resources and pushed towards privatisation and state divestiture. The greater percentage of Ghana's mineral resources is gold, from which the country obtained its previous name i.e., Gold Coast. Gold occurrences, as well as mining operations in Ghana, are predominantly concentrated within the Paleoproterozoic Birimian greenstone belts, approximately 10 km–15 km wide, separated by deep sedimentary basins [1–3]. It is noteworthy that, gold occurrence in Ghana is largely structurally controlled within the Paleoproterozoic Birimian greenstone belts [4]. Despite the available geological information, exploration success has been difficult even in the most prospective gold provinces in Ghana [5]. This lack of gold exploration successes has been attributed to many factors, such as; depletion of the easy-to-find mineral deposits [6] and the improper understanding of the complexities of the regolith concealing the underlying mineralisation from detection [7]. As a response to the aforementioned difficulties associated with exploration successes, mineral exploration geoscientists in recent years have employed various data integration methods as well as geographic information systems (GIS). The use of these data integration methods, as well as GIS in the mineral exploration procedure, is to determine and evaluate the influence of diversely derived mineral predictors sourced from various geoscientific datasets that are to be synthesised into a mineral potential map (MPM). This MPM procedure, though is able to outline promising target zones for mineral occurrence is a very cumbersome task. Given this, various data integration techniques that have been developed empirically (data-driven) or conceptually (knowledge-driven) have been employed towards the generation of mineral prospectivity models [8].

For the empirically-based data integration techniques, the weight (influence) of each of the mineral predictors are determined by assessing their spatial correlation with the existing locations of the targeted mineral resource. This type of data-integration technique is dependent on adequate availability of known occurrence locations of the targeted mineral resource (known as brown fields). Examples of the empirically-based data integration techniques are frequency ratio [9,10], Shannon entropy [11], multi-index overlay [12], evidence belief functions [13] and machine learning methods [14–16]. In the case of the conceptually-based data integration techniques, the weight of the mineral predictors to be synthesised are determined using expert judgment. These techniques are recommended for under-explored virgin areas where little or no location of the sought-after mineral deposit can be found.

There are two modes of data integration techniques for the conceptually-based data integration MCDM method. The first mode is the conceptually-based data integration technique that aims to determine the weight or influence of various predictors towards the generation of an MPM. This technique is commonly used in geoscience predictive modeling and includes analytical hierarchy process (AHP) as well as fuzzy AHP [17] and best worst method [8]. In the case of the second mode of conceptually-based data integration techniques, various classes (alternatives) within the selected mineral predictors are prioritised and ranked before synthesising them into an MPM. Commonly employed examples in geoscience are analytical network process [18], ELECTREE [19], PROMETHEE [20], TOPSIS [21] and VIKOR [22] method.

For several studies in geoscience geared towards the production of a predictive model by utilising multicriteria decision methods (MCDM), various predictors are integrated by employing either the first or second mode of the two aforesaid modes of the conceptually-based data integration techniques [8,23]. It is uncommon to find a combination of the various modes of conceptually-based data integration techniques used in the generation of MPM in literature. However, when used together, they can help determine the influence of each predictor as well as prioritise and rank the different alternatives within each predictor. Thus, the combined usage of these two modes of conceptually-based data integration techniques is considered to be more effective in predictive modeling as compared to using them separately [24,25]. For this reason, this study has adopted the best-worst method and the Technique for Order of Preference by Similarity (TOPSIS) methods to respectively determine the weights of nine predictors sourced from geophysical and geological datasets as well as rank the alternatives that characterise each of these predictors. The best-worst methods, though a recently developed MCDM, has been essential in determining the influence of various predictors that are to be synthesised into an MPM. Also, the TOPSIS technique, as an MCDM has shown prowess towards the prioritisation and ranking of various alternatives of predictors by determining their closeness to an ideal solution. A combination of these two methods (BWM and TOPSIS) can thus be essential in generating optimal results. Given this, this study's overarching objective is to synthesise mineral-related predictors sourced from geological, magnetic and radiometric datasets acquired over the Abansuoso Area of the Ahafoano North District of Ghana's Ashanti Region to develop an MPM based on a proposed BWM-TOPSIS hybrid procedure. The prediction-area plot and receiver operating characteristics curve would be employed to evaluate the performance of the BWM-TOPSIS based MPM.

2. Study area and geological setting

The study area falls within the Abansuoso Area, which is subsequently situated in the Ahafoano North District of Ghana's Ashanti Region. It has an area size of approximately 590.24 km². The study area is bounded by coordinates 738440 mN, 763240 mN, 563440 mE and 587240 mE within the WGS 84 of UTM Zone 30 north.

2.1. Geological setting

In Fig. 1(a), the Baoule Mossi domain is seen to occupy the southeastern part of the West African Craton, an area dating to the Paleoproterozoic [26]. During the Eburnean orogeny, the formation of the Birimian terrane is deemed to have occurred at 2.1 Ga [27,

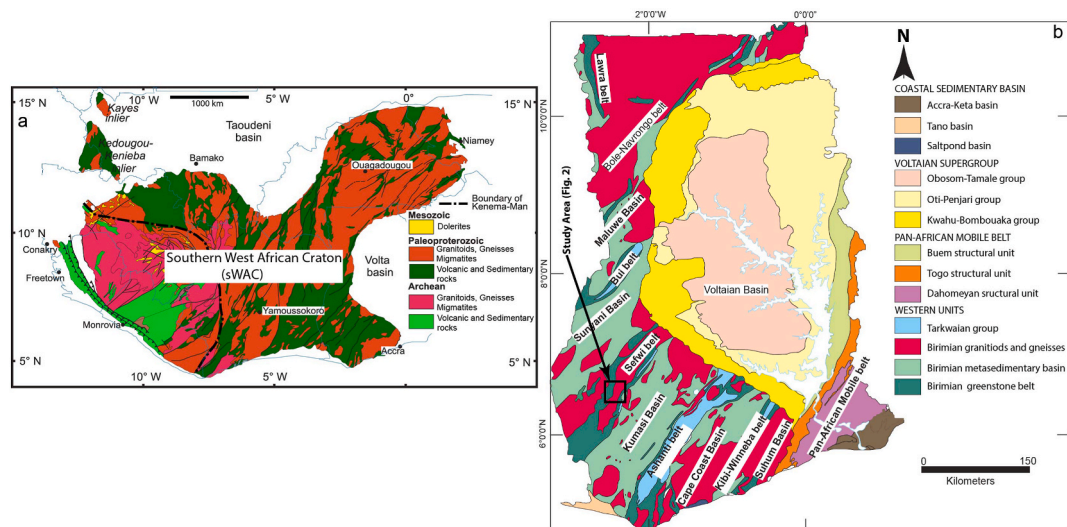


Fig. 1. Map of (a) the West African Craton, (b) Ghana showing the Birimian greenstone gold belts (modified after Allibone et al. [40]). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

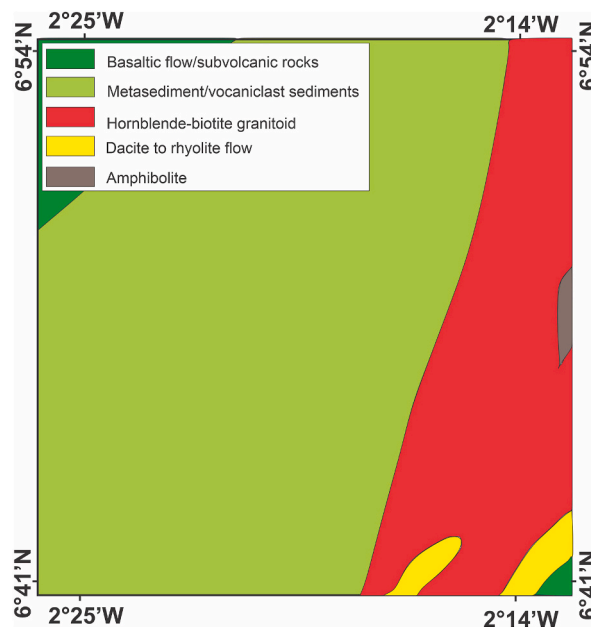


Fig. 2. Map of the study area characterising the main lithological units.

[28,29]. This orogeny produced several nearly evenly spaced greenstone gold belts, separated by metasedimentary basins [30]. The greenstone gold belts, and metasedimentary basins have been intruded by granitoids emplaced at different times [31,29,32]. In Ghana, there are six greenstone gold belts, which comprises Lawra, Bole-Nangodi, Bui, Sefwi, Ashanti and Kibi-Winneba belts from the northwest to southeast as shown in Fig. 1(b) [33]. For these aforementioned greenstone gold belts, only two (Ashanti and Sefwi) have within them operationally-productive activities by various mining firms. Advanced exploration programmes as well as minimal mining operations are carried out within the other four belts. It is noteworthy that the Maluwe, Sunyani, Kumasi and Suhum-Cape Coast basins make up the metasedimentary basins in Ghana. The greenstone gold belts share similar lithological and structural features [28,30]. They are generally made up of tholeiitic, calc-alkaline massive and pillow lavas, subvolcanic rocks now metamorphosed under greenschist facies and amphibolite facies conditions [34,35]. The metasediments are chiefly argillites and wackes that have also undergone metamorphism through the greenschist to amphibolite facies conditions [30,36–38]. Four types of granitoids intrude the belts and basins and they comprise the Cape Coast, Dixcove, Winneba and Bongo type. The granitoids generally show TTG (tonalite-trondhjemite-granodiorite) compositional features, I-type and dominantly metaluminous to weakly peraluminous signatures [31].

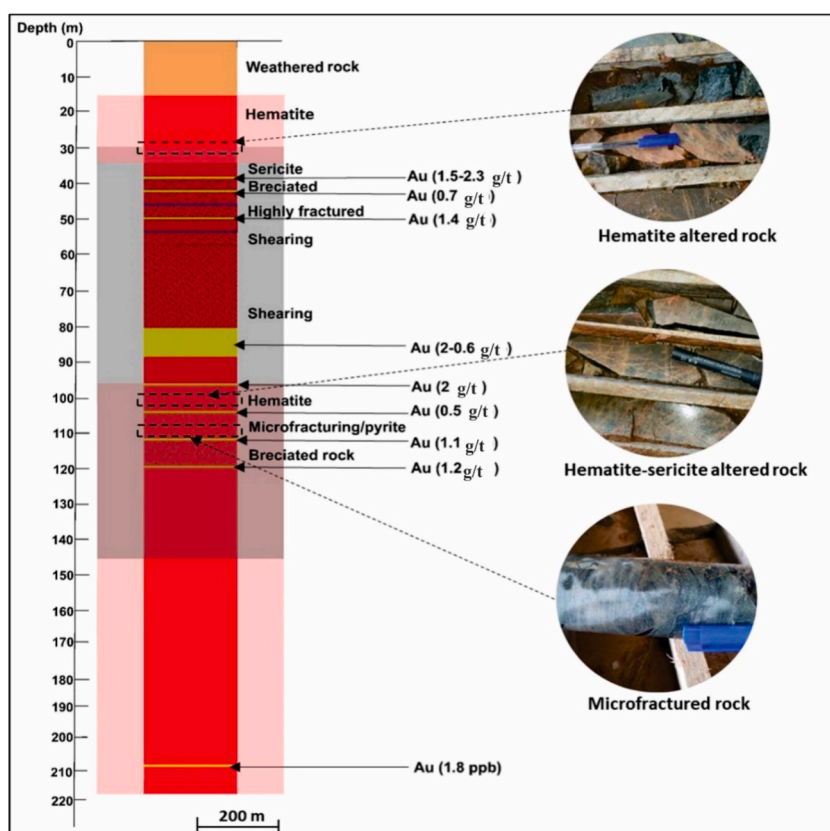


Fig. 3. Schematic representation of drill core with photographs of hematite altered rock, brecciated sericite altered rock, and hair-like fractures filled with hematite.

For the Bongo type, it shows adakitic characteristics [30]. The study area falls within the Sefwi gold belt in the Abansuoso area (Fig. 2). According to Perrouy et al. [37], this belt hosts varying grades of gold mineralisation. The Sefwi greenstone gold belt lies northwest of the Ashanti Belt and is underlain by metasedimentary, metavolcanic and volcano-sedimentary rocks with observed intrusions from granitoids and mafic units similar to the other Birimian greenstone gold belts of Ghana. The eastern and western portions of this belt are respectively bordered by the Kumasi Basin and Sunyani Basin [4,39]. Within the Sefwi belt, andesite, basalts, dacites, and rhyolites of varied metamorphic grades are predominantly observed together with volcanoclastic units (composed of epiclastics, pyroclastics and schists). Syn-volcanic tonalitic to granodioritic granitoids and minor sedimentary rocks are also found within the Sefwi belt [38]. These rocks have undergone metamorphism under epidote-amphibolite to lower amphibolite facies condition at pressure and temperature of 4–6 kbar and 490 °C - 580 °C. The central part of the belt has experienced mostly amphibolite facies metamorphism, whereas the southeastern part has undergone epidote-amphibole facies metamorphism [39].

2.2. Mineralisation style of the area

The study area falls within the Sefwi greenstone gold belt in the Abansuoso area (Fig. 2). This belt hosts varying grades of gold mineralisation [37]. It has been found that out of the six greenstone gold belts, two are the most productive, and the others are being explored and drilled to a minimal extent. Some of the major gold deposits in Ghana are Obuasi, Konongo, Wassa, Damang, and Tarkwa Gold. These deposits are located along the NE trending faults, which are part of the regional structural architecture [40]. The gold in these deposits is related to dispersed gold-containing arsenopyrites that can be found in quartz veins and stockworks along with free gold [40]. Polymetallic sulphides such as pyrrhotite, pyrite, chalcopyrite, and galena are also present. Located northwest of the Ashanti belt, the Sefwi greenstone gold belt consists of metavolcanic, metasedimentary, and volcano-sedimentary rocks intruded by mafic and granitic units, similar to the other Birimian greenstone belt in Ghana [34,39]. The primary structural feature that controls mineralisation in the study area is the Sefwi thrust fault. The mineralisation in the study area occurs in shear zones linked to the thrust fault, similar to what is observed in other gold deposits [40,41]. With a brittle component, the shear zone is ductile-sinistral. Several phases of metamorphism and deformation have caused the sheared biotite gneiss of intermediate composition and the variable composition schist to host the mineralisation [3]. Akin to other gold deposit of Ghana, arsenopyrite and, in some samples, a minor amount of pyrite constitutes the majority of the main ore minerals; both are found in the mineralised zones. Arsenopyrite, pyrite, and galena respectively make up 50 %, 30 %, and 20 % by volume of the mineralised rocks. Sizes of the two major sulphide ore minerals can range from 1

to 8 m. Typically, gold mineralisation occurs in free form in sericite altered host lithologies, as well as in micron-sized gold inclusions associated with euhedral arsenopyrite and pyrite [41]. When gold is found associated with sericite alteration; dolomite is also present. Thus, pyrite and arsenopyrite are found widely distributed. Salinity-wise, the gold bearing fluids are characteristically low. However, the associated $CO_2 - N_2 \pm CH_4$ fluids observed within the gold-bearing fluids makes it strongly affinitized to CO_2 . Also included in these gold-bearing fluids are aqueous $H_2O-NaCl$ and H_2O-CO_2-NaCl fluids. The alteration sequence within the area's wall rock infiltrates the materials via stringers along shear-induced weak zones. Pyrite with hematite inclusions as well as quartz-sericite are the predominant alteration minerals within the study area, whose associated gold grade have been documented. Sericite alteration associated with fine-grained dolomite and pyrite distinguishes the principal mineralised zones. It has been determined that the preliminary alteration sequence are chlorite-carbonate-hematite-sericite-quartz/pyrite with minor amount of albite [3]. Fracture planes and traces of hematite alteration transition into sericite alteration within brecciated fragments of mildly weathered schist define the mineralised zone. At other sections, faintly cataclarised, sericite-altered brecciated pyrite schist is observed (Fig. 3).

3. Materials and methods

3.1. Theory of the hybrid BWM-TOPSIS approach

By employing the proposed hybrid BWM-TOPSIS approach in this study for the integration of the geoscientific mineral influencing factors (criteria), the BWM was employed in computing the criteria weights upon the attainment of an acceptable consistency ratio for the pairwise comparison matrix (PCM) created. Afterwards, the TOPSIS method was applied on the various alternatives within each criteria to prioritise and rank them. The theoretical background of the BWM and TOPSIS techniques have been described in subsections 3.1.1 and 3.1.2.

3.1.1. BWM

The BWM technique was put forward by Rezaei [42] to be used in addressing multicriteria decision making (MCDM) problems. In dealing with MCDM issues of mineral prospecting, the BWM technique computes the weight for various mineral influencing factors (criteria) as well as each factor's classes (alternatives) on the basis of a pairwise comparison.

Paramount among the reasons why decision-makers use the BWM in dealing with mineral-related MCDM problems comprises (1) the pairwise comparisons (PC) created during the use of BWM technique is lesser than the PC created for other MCDM methods such as the analytical hierarchy process (AHP) technique, (2) PC created for various mineral influencing factors require only positive integers, (3) BWM-based specified weight coefficients are more efficient and (4) the BWM can also be synthesised with other MCDM techniques to compute weights for criteria. The utilisation of the BWM for the calculation of weights of various factors dwells on the following five stages [42,43]:

- Decision makers first outline a set of mineral influencing factors $\{F_1, F_2, \dots, F_n\}$ that the decision-maker would use in the attainment of the set objective (mineral potential mapping in this case).
- A decision on which of the mineral influencing factors is the best (highly beneficial criteria) and worst (least beneficial criteria) is taken by the decision maker. It is noteworthy that, this stage is devoid of comparison among the mineral influencing layers.
- By assigning a value in the range of 1–9, decision makers outline their preference for the highly beneficial criteria with respect to the other criteria. This generates a row vector known as the best-to-others vector (F_B) shown in equation (1).

$$F_B = (f_{B1}, f_{B2}, f_{B3}, \dots, f_{Bn}) \quad (1)$$

f_{Bn} depicts the importance of the most beneficial criteria (B) over another criteria n . Thus, $f_{BB} = 1$.

- By assigning a value in the range of 1–9, decision makers outline the preference of all the other mineral influencing factors over the least beneficial criteria. The pairwise comparison created is a column vector known as the others-to-worst (F_W) expressed in equation (2).

$$F_W = (f_{1W}, f_{2W}, f_{3W}, \dots, f_{nW}) \quad (2)$$

f_{nW} characterises the comparative importance of a criterion n over the criterion with the least beneficial attribute (W). Hence, $f_{WW} = 1$.

- The optimal weights ($W_1^*, W_2^*, W_3^*, \dots, W_n^*$) and ξ^* (uncertainty parameter) are computed by solving the optimisation problem (equations (3a) and (3b)).

$$\min \xi \text{ s.t for all } j, \left| \frac{W_B}{W_j} - a_{Bj} \right| \leq \xi \quad (3a)$$

$$\left| \frac{W_j}{W_W} - a_{jW} \right| \leq \xi \sum_j W_j = 1, W_j \geq 0 \quad (3b)$$

The computed value for the ξ^* and their respective consistency index (CI) in Table 1 can be used in equation (4) to calculate for the

Table 1

The consistency Indices (CI) proposed by Rezaei (2015).

a_{BW}	1	2	3	4	5	6	7	8	9
Consistency Index ($\max \xi$)	0.00	0.44	1.00	1.63	2.30	3.00	3.73	4.47	5.23

consistency ratio (CR).

$$CR = \frac{\xi^*}{CI} \quad (4)$$

CR values typically range from 0 to 1. CR values close to 1 denote high inconsistencies whereas values close to 0 depict high consistencies. Thus, during decision making based on the BWM, pairwise comparisons created with CR values close to 0 are the most preferred [42].

3.1.2. TOPSIS method

Through a research conducted by Hwang and Yoon [44], the TOPSIS method was first proposed. This aforementioned method was developed further through studies carried out by Lai et al. [45] and Yoon and Hwang [46] to be used as an MCDM technique capable of ranking alternatives (classes) within variously identified mineral influencing factors (criteria). In the implementation of the TOPSIS method, the highest or lowest ranked alternative (class) is the class with the highest or lowest relative closeness towards the ideal solution. Implementation procedures associated with the TOPSIS method starts with the creation of a decision matrix (equation (5)) for the influencing factors whose alternatives are to be ranked.

$$\begin{array}{c}
 AL_1 \\
 AL_2 \\
 \vdots \\
 AL_i \\
 \vdots \\
 AL_m
 \end{array}
 \begin{array}{c}
 F_1 \quad F_2 \quad \cdots \quad F_j \quad \cdots \quad F_n \\
 \left(\begin{array}{cccccc}
 f_{11} & f_{12} & \cdots & f_{1j} & \cdots & f_{1n} \\
 f_{21} & f_{22} & \cdots & f_{2j} & \cdots & f_{2n} \\
 \vdots & \vdots & & \vdots & \vdots & \vdots \\
 f_{i1} & f_{i2} & \cdots & f_{ij} & \cdots & f_{in} \\
 \vdots & \vdots & & \vdots & \vdots & \vdots \\
 f_{m1} & f_{m2} & \cdots & f_{mj} & \cdots & f_{mn}
 \end{array} \right)
 \end{array} \quad (5)$$

From equation (5), F_j represents the j th influencing factor or criteria ($j = 1, 2, 3, \dots, m$); the i alternatives (classes) are depicted as AL_i , where $i = 1, 2, 3, \dots, n$; f_{ij} is of crisp value in nature and denotes the performance rating of each alternative (AL_i) and their respective criterion (TL_j). The decision matrix (in equation (5)) created is then normalised as shown in equation (6).

$$r_{ij} = \frac{f_{ij}}{\sqrt{\sum_{j=1}^n f_{ij}^2}} \quad (6)$$

where r_{ij} is the normalised decision matrix

Through the expression captured in equation (7), further computations are carried out to obtain the weighted normalised decision matrix (e_{ij}).

$$e_{ij} = w_j r_{ij}; i = 1, 2, 3, \dots, n; j = 1, 2, 3, \dots, m \quad (7)$$

w_j in equation (7) represents the weight of the j th influencing factor (criteria) and thus $\sum_{j=1}^n w_j = 1$.

Afterwards, the positive (E^+) and negative (E^-) ideal solutions are determined as shown in equations (8a) and (8b).

$$E^+ = \{e_1^+, \dots, e_n^+\} = \left\{ \left(\max_i e_{ij} \mid j \in J \right), \left(\min_i e_{ij} \mid j \in J' \right) \right\} \quad (8a)$$

$$E^- = \{e_1^-, \dots, e_n^-\} = \left\{ \left(\min_i e_{ij} \mid j \in J \right), \left(\max_i e_{ij} \mid j \in J' \right) \right\} \quad (8b)$$

From equation (8), J and J' respectively depicts a collection of beneficial and non-beneficial criteria.

By employing the Euclidean distance approach with n -dimension, each alternative's closeness with respect to the positive and negative ideal solutions is expressed as shown in equations (9a) and (9b).

$$S_i^+ = \sqrt{\sum_{j=1}^n (e_{ij} - e_j^+)^2}; i = 1, 2, 3, \dots, m \quad (9a)$$

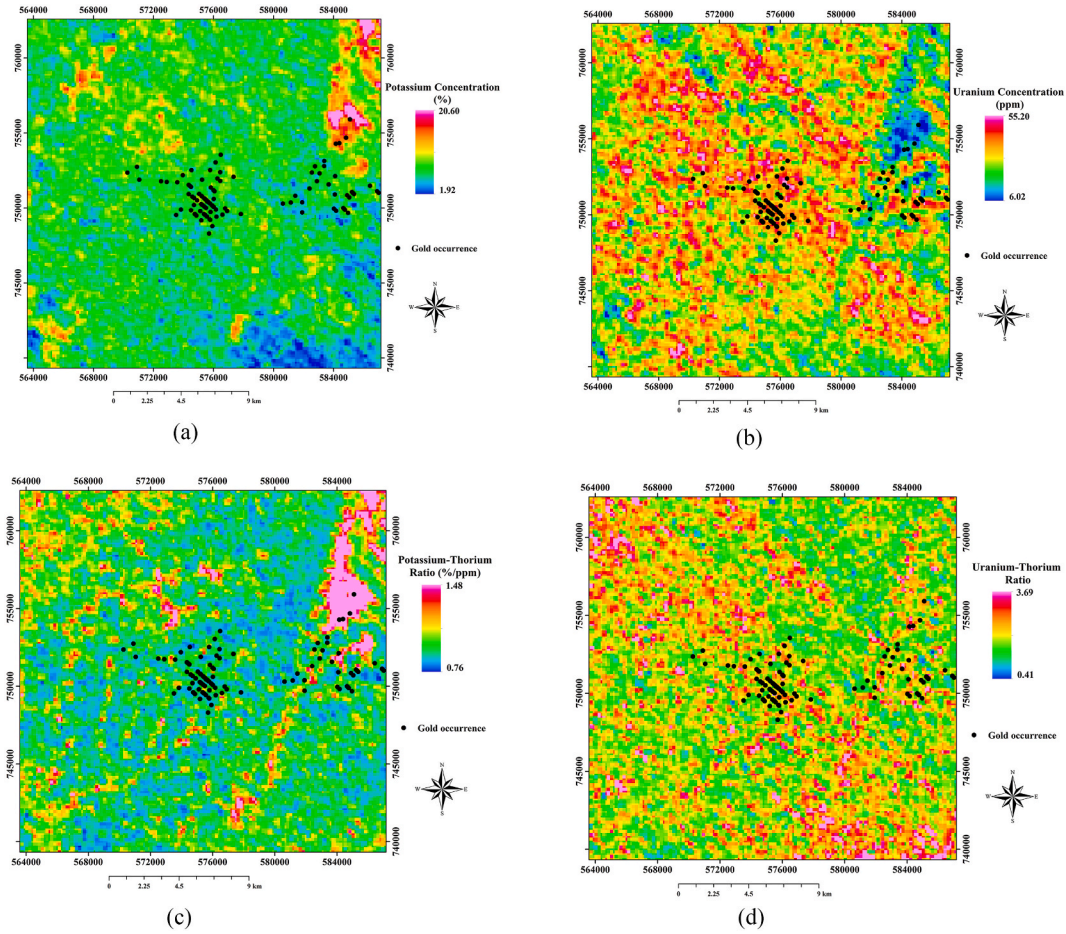


Fig. 4. Map of (a) Potassium concentration factor (b) Uranium concentration factor (c) Potassium-thorium factor (d) Uranium-thorium factor.

$$S_i^- = \sqrt{\sum_{j=1}^n (e_{ij} - e_j^-)^2}; i = 1, 2, 3, \dots, m \quad (9b)$$

For each alternative A_i , the relative closeness (RC) in relation to a positive ideal solution (E^+) is calculated as shown in equation (10).

$$RC = \frac{S_i^-}{(S_i^- + S_i^+)}; i = 1, 2, 3, \dots, m \quad (10)$$

The RC values obtained for each alternative is essential in ranking them. Thus, alternatives with higher RC values are deemed better alternatives than those with lower RC values.

3.2. Sources and descriptions of datasets used

Unequivocally, the fundamental stage in mineral prospectivity modeling that cannot be overemphasised is the identification and collection of suitable geoscientific datasets, from which factors with significant relevance towards the sought-after mineral deposit. Accordingly, the geoscientific datasets were identified and selected for this present study by taking into cognisance the availability of the aforesaid dataset in tandem with the theoretical and empirical basis of the spatial correlation between the datasets and gold mineralisation occurrence within the study area. Hereupon, nine mineral influencing factors comprising analytic signal, uranium (eU) concentration, reduction-to-equator, lineament density, potassium (K) concentration, potassium-thorium (K-eTh) ratio, uranium-thorium (eU-eTh) ratio, distance to faults and geology were propounded to delineate the prospective zones of gold occurrence within the Abansuoso Area. In this study, the said mineral influencing factors were derived from airborne magnetic and radiometric datasets as well as geology vector shapefiles sourced from the Ghana Geological Survey Authority (GGSA). A description of the basis upon which these mineral influencing factors were identified and selected from these three data sources have been explained in this section. The mineral influencing factors derived from the radiometric data generally characterise the radioelement (K, eTh and eU)

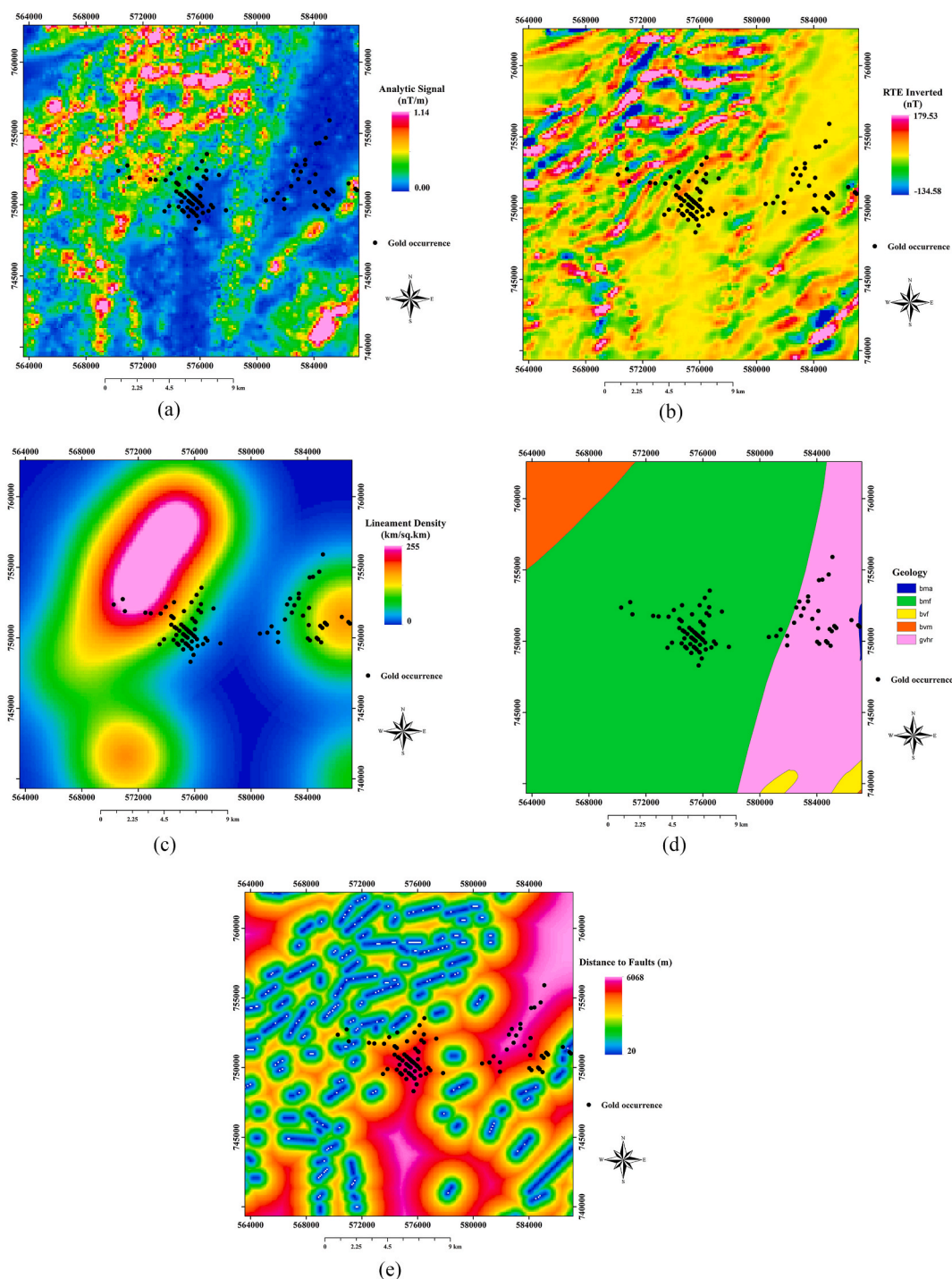


Fig. 5. Map of (a) Analytic signal factor (b) Reduction-to-equator Inverted factor (c) Lineament density factor (d) Geology factor (e) Distance-to-faults factor.

concentration distribution over a region of interest [8]. The radiometric ratio maps (K/eTh , eU/eTh) particularly are undoubtedly significant in outlining mineralised hydrothermal zones as well as alteration zones that are relevant towards gold mineralisation occurrence within the Abansuoso Area and its related geological terranes [3,12]. Thus, four radiometric-sourced layers comprising K (Fig. 4(a)), eU (Fig. 4(b)), K/eTh (Fig. 4(c)) and eU/eTh (Fig. 4(d)) were identified as important factors in this study for the development of a model that outlines the potential zones of mineral occurrences. It is worth noting that the gridding and processing of these geophysically-sourced factors (from magnetic and radiometric) were implemented using the Geosoft Oasis Montaj software. In the case

Table 2
Class scores for the symbols in the geology factor.

Influencing Factor	Geological Symbol	Class Score
Geology	gvhr	5
	bmf	3
	bvm	5
	bvf	1
	bma	3

of the magnetically-derived factors; the analytic signal factor characterising the distribution of magnetic intensity gradient over an area, was employed on the magnetic intensity grid. In low magnetic latitudes, asymmetric problems are observed between the magnetic responses and the source which produces those anomalies [17,47]. By making the observed magnetic responses and the source producing them symmetric, the reduction-to-equator was employed and was further inverted. Thus, the RTE Inverted factor characterises the distribution of magnetic intensities over the study area. Regions with high magnetic intensities, as exhibited by analytic signal factor (Fig. 5(a)) and RTE Inverted factor (Fig. 5(b)) could be associated with sulphide-associated indicator minerals such as pyrite, arsenopyrite as well as magnetite. Also, from the magnetic data, the lineament density factor was generated by employing the CET (Centre for Exploration Targeting) grid analysis technique [9,47–49]. Lineaments such as faults, fractures, etc., depict low pressure zones and consequently act as conduits for hosting various mineralised fluids. In view of this, regions characterised by high lineament density (Fig. 5(c)) give an indication of intense fracturing occurrence over the study area. These geophysically-derived factors were exported to a GIS environment in a raster format. The geological factor was also converted from a polygon shapefile to a raster format in a GIS environment. Mineralisation occurrences have been observed within close distance to major structures such as faults, fractures, folds, etc. This further suggests that a decreased distance to major fault zones can be earmarked as highly prospective zones of mineralisation occurrences within an area [50]. Thus, for the distance to fault factor (shown in Fig. 5(e)), regions that are distal from faults could characterise the low prospective zones and vice versa. The geology factor (Fig. 5(d)), which was sourced from the 1:1,000,000 digital geological map compiled by Agyei Duodu et al. [33], was obtained in vector format and comprised various lithological units (bma, bmf, bvf, bvm and gvhr) within the study area. The bma unit is predominantly characterised by amphibolite facies whereas the bmf lithological unit comprises mainly metasediments and volcanoclastic sediments. For the bvf, its terrane is generally characterised by dacite to rhyolite flow. In the case of the bvm, its unit is dominated by basaltic flow and subvolcanic rocks whereas hornblende-biotite granitoids are observed to characterise the gvhr lithological unit. All except one (geology factor) of the nine influencing factors identified and employed for the study had their pixels (alternatives) values represented by numbers. For the geology factor, it alternatives were represented by their geological symbols (string format), as shown in Fig. 5(d). In order to efficiently rank all the pixels (alternatives) within each of the nine influencing factors using the TOPSIS method, the geological symbols were replaced by numbers, taking into cognisance their significance in relation to gold occurrence within the Abansuoso Area. Thus, the lithological class with the highest influence towards gold occurrence was given a score of 5, whereas the other classes deemed to have a comparatively lower relevance to gold occurrence were assigned values less than 5, as shown in Table 2. All these nine mineral influencing factors were rescaled to a spatial resolution of 100 m, resulting in the generation of 60,491-pixel counts for the study area.

Geospatial-based predictive models produced cannot be meaningfully considered for any form of analysis until they have been evaluated and validated [51]. Subsequently in this study, the existing locations of gold occurrences (target mineral) observed were utilised to assess the performance of the predictive model to be generated in this study. This occurrence data comprised 99 anomalously high Au locations in ppm (parts per million) and zones of artisanal mining operations identified in the study area. Au geochemical anomalies were in point shapefile format together with their respective geographic coordinates (longitudes and latitudes) as well as the Au geochemical anomaly channel in ppm. This Au geochemical sample point data was acquired during a sampling field work in 2020. For the locations of artisanal mining operations (known as galamsey in our local setting), locations were identified using the GPS (geographic positioning system).

3.3. Evaluation and validation of hybrid BWM-TOPSIS model

Through the implementation of the prediction-area plot and the receiver operating characteristics (ROC) curve techniques, the efficacy of the model that would be developed by employing the hybrid BWM-TOPSIS approach was assessed.

3.3.1. Prediction-area plot

Utilising the existing locations of Au occurrences identified in the study area, the accuracy and reliability of the mineral potential map developed based on hybrid BWM-TOPSIS approach for the Abansuoso Area was evaluated. Given this, the 99 known Au occurrence points were overlaid on the BWM-TOPSIS based MPM (which has been discretised by applying the natural breaking classification method) in GIS environment. For each of the classes obtained for the MPM generated, their corresponding study area percentage (fraction) was determined by dividing their respective class coverage by the total area coverage of the MPM produced (known as fraction of area). Also, the fraction of occurrence of the Au mineral for each class was determined by dividing the Au occurrences observed in each class by all the Au occurrences observed within the study area. Hence, the predictive capacity for each class within the BWM-TOPSIS based MPM generated was expressed in terms of the aforesaid fractions [10,52]. By plotting the fraction

Table 3

Best-to-others comparison for the three DMs based on assignment values proposed by Rezaei [42].

Criteria	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9
DM1: Best Criterion - F_3	2	2	1	4	3	7	5	3	5
DM2: Best Criterion - F_1	1	4	2	3	3	6	5	4	7
DM1: Best Criterion - F_3	3	3	1	3	2	9	5	4	5

Table 4

Others-to-worst comparison for the three DMs based on assignment values proposed by Razaei (2015).

Criteria	DM1: Worst Criterion - F_6	DM2: Worst Criterion - F_9	DM3: Worst Criterion - F_6
F_1	7	8	5
F_2	5	5	4
F_3	9	7	8
F_4	4	6	6
F_5	4	6	7
F_6	1	2	1
F_7	4	3	4
F_8	5	5	5
F_9	4	1	4

of area as well as the fraction of Au occurrence concurrently (both on separate y-axis) against the prospectivity score of the classes obtained for the MPM (on the x-axis), the possibility of identifying an undiscovered Au deposit is smaller for a spatially large predictive map than a spatially small predictive map for two maps that are spatially distinct with unequal areal extent is determined [52]. For this study, the aforementioned plot produced represents the P-A plot that evaluates the efficacy of the hybrid BWM-TOPSIS based mineral prospectivity model. This plot generated comprises two curves; fraction of study area for each discretised class and the fraction of Au occurrence for each discretised class.

3.3.2. Receiver operating characteristics (ROC) curve

The implementation of the ROC in this study was essentially carried out to assess the efficacy of the predictive model generated by using the hybrid BWM-TOPSIS method because its assessment capability is universal. Moreover, the ROC is very easy to employ. Thus in this study, the 99 known Au occurrence points were carried out to determine the performance of the MPM to be developed. An ROC outputs a curve that is usually characterised by an x-axis, which depicts the false positive rate (the probability that a region is predicted to be positive while in actual sense, it is not positive) and a y-axis that represent true positive rate (the probability that a positively predicted region is indeed positive). The area under ROC curve (AUC) produces a score that typically ranges from 0.5 to 1.0. AUC scores closer to 0.5 are deemed to demonstrate a weak model performance whereas scores closer to 1.0 exhibit optimum model performance. In general, predictive models with AUC scores greater or equal to 0.7 are recommended and are considered to be good models [53,54].

4. Results

4.1. Application of the BWM-TOPSIS model

From the BWM-TOPSIS model outlined under section 3.1, the comparative importance (weight) of the nine mineral influencing factors were determined using the BWM technique as demonstrated under section 3.1.1. Application of the TOPSIS method in prioritising and ranking the pixels (alternatives) within the nine factors has been demonstrated based on the procedures outlined under section 3.1.2.

4.1.1. Weighting of mineral influencing factors by the BWM technique

The assignment of weights to the nine mineral influencing factors based on the BWM technique was carried out by decision makers (three in number, with expertise in mineral prospecting) to determine the importance of each of the nine factors in relation to gold occurrence within the Abansuoso Area. For each decision maker, criteria numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8$ and F_9 were used to denote each of the nine factors that respectively comprise geology, distance to faults, lineament density, analytic signal, RTE Inverted, K, eU, K-eTh ratio and eU-eTh ratio. The first and the third decision makers (DM_1 and DM_3) selected the best and the worst influencing factor to be, respectively, the lineament density (F_3) and potassium (F_6). The geology (F_1) and the uranium-thorium ratio (F_9) factors were chosen as the best and worst influencing factors (criteria) respectively by the second decision maker (DM_2 as shown in Tables 3 and 4). By solving the optimisation problem expressed in equation (3), weights were assigned to each mineral influencing factor based on each decision maker. These weights assigned are deemed reliable only when their respective consistency ratios (CR) are less than 0.1 as expressed in equation (4). Thus, weights obtained (shown in Table 5) were observed to have computed CR values of 0.0102, 0.0126 and 0.0104 for the first, second and third decision maker respectively. These weights obtained for each mineral influencing factor by

Table 5
BWM-based computed criteria weights.

Criteria Symbol and Name	Computed Criteria Weights (W^*)			Sum of DM Weights	Mean Weights
	DM1	DM2	DM3		
F_1 (Geology)	0.1545	0.2756	0.1796	0.5388	0.1796
F_2 (Distance to Faults)	0.1545	0.0854	0.1087	0.3486	0.1162
F_3 (Lineament Density)	0.2554	0.1708	0.2717	0.6979	0.2326
F_4 (Analytic signal)	0.0772	0.1138	0.1087	0.2998	0.0999
F_5 (RTE Inverted)	0.1030	0.1138	0.1630	0.3799	0.1266
F_6 (Potassium)	0.0288	0.0569	0.0272	0.1129	0.0376
F_7 (Uranium)	0.0618	0.0683	0.0652	0.1953	0.0651
F_8 (K-eTh ratio)	0.1030	0.0854	0.0815	0.2699	0.0900
F_9 (eU-eTh ratio)	0.0618	0.0300	0.0652	0.1570	0.0523
ξ^*	0.0536	0.0659	0.0543		
CR	0.0102	0.0126	0.0104		

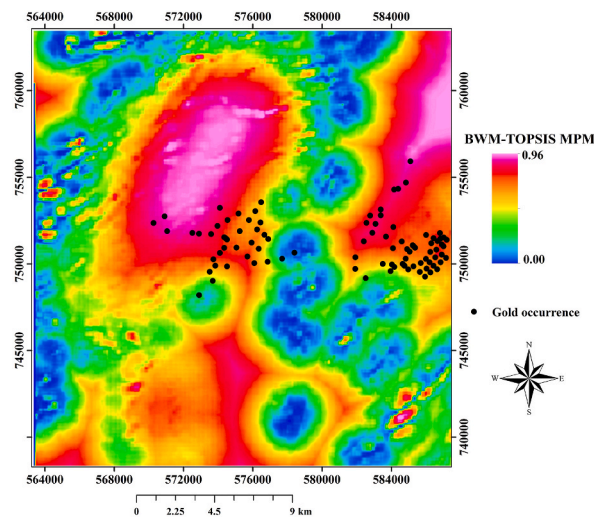


Fig. 6. MPM produced based on the BWM-TOPSIS procedure.

each of the decision makers were summed up and divided by the number of decision makers (3 in this study) to obtain the mean weight for each of the factors (shown in Table 5). From the results obtained for the mean weight of the influencing factors in Table 5, the influencing factors were ranked from the lowly to the highly relevant criteria as potassium (F_6 : Mean weight = 0.0376), eU-eTh ratio (F_9 : Mean weight = 0.0523), Uranium (F_7 : Mean weight = 0.0651), K-eTh ratio (F_8 : Mean weight = 0.0900), analytic signal (F_4 : Mean weight = 0.0999), distance to faults (F_2 : Mean weight = 0.1162), reduction-to-equator inverted (F_5 : Mean weight = 0.1266), geology (F_1 : Mean weight = 0.1796) and the lineament density (F_3 : Mean weight = 0.2326).

4.1.2. Pixels (alternatives) prioritisation and ranking using the TOPSIS method

After employing the best-worst method technique to determine the comparative importance for each of the nine mineral influencing factors over the Abansuoso Area, there was the need to prioritise and rank the alternatives (pixels) characterising the aforementioned study area; and this was carried out using the TOPSIS technique. In order to successfully prioritise and rank the alternatives covering the Abansuoso Area, a matrix comprising $m = 60,491$ rows (which denotes the number of study area pixels) and $n = 9$ (denoting the number of mineral influencing factors) as expressed in equation (5) was first created. In accordance with the procedures outline in section 3.1.2., the use of the TOPSIS technique to prioritise and rank the pixels within the aforesaid factors were implemented based on the expressions captured as equations (6), (7) and (8a), 8b, 9a, 9b and 10. Considering the results obtained based on the beneficial and non-beneficial criteria (influencing factors) selected, a weighted normalised grid was generated for each factor based on the implementation of the normalised weighted matrix produced using equation (7). These normalised weighted grids (obtained from the normalised weighted matrix) were essential in determining the positive and negative ideal solutions as well as the relative closeness (RC) for each of the alternatives.

4.2. BWM-TOPSIS based MPM

Considering the relative closeness values obtained for the 60,491 ranked pixels (alternatives), a grid characterising the prospective

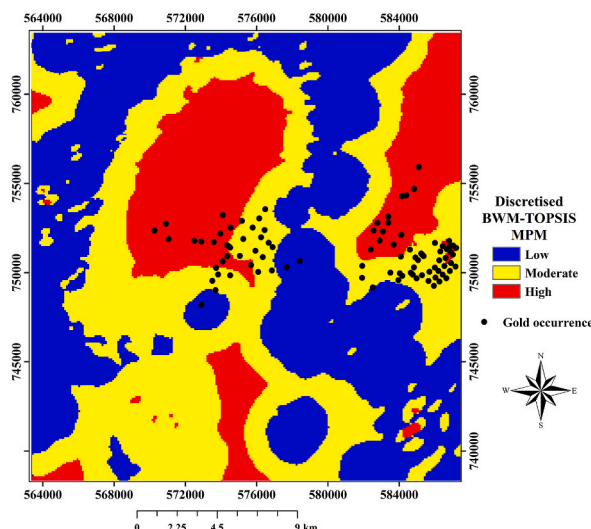


Fig. 7. Discretised Map of the MPM produced based on the BWM-TOPSIS procedure.

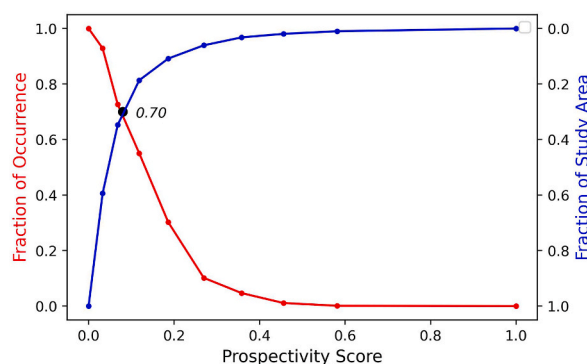


Fig. 8. Prediction-area plot for the MPM produced based on the BWM-TOPSIS procedure.

zones of mineral occurrence over the Abansuoso Area was produced based on the BWM-TOPSIS approach. The MPM generated based on the BWM-TOPSIS approach was implemented by taking into account the cell size of each alternative (pixel); which was $100 \text{ m} \times 100 \text{ m}$ in this study on the basis of their respective RC values obtained. From the MPM generated (shown in Fig. 6), regions deemed to be prospectively high in terms of mineral occurrence were associated with higher relative closeness values (prospectivity values close to 0.96) whereas regions characterised by low mineral potential have prospectivity values close to 0.00. It was observed that, regions within the metavolcanic and the volcanoclastic sediment dominated terrane, with intense lineation (due to the existence of fracture and shear zones along the north-central portion of the area) were delineated to be of moderate to high potential. This corroborates with literature as gold occurrence in Abansuoso Area is mostly observed along strongly sheared zones [3]. The MPM developed was further discretised into three classes using the natural breaking classification method into low, moderate and high prospectivity classes as shown in Fig. 7. These aforementioned prospectivity classes covered areal extent of 237.13 km^2 , 225.66 km^2 and 127.45 km^2 respectively.

4.3. Evaluation of the BWM-TOPSIS based MPM

By utilising the ROC curve and the P-A plot model evaluation approaches, the performance of the mineral potential map developed over the Abansuoso Area based on the hybrid BWM-TOPSIS approach was evaluated. Outputs obtained from utilising the P-A plot (see Fig. 8) shows the intersection point between the fraction of mineral occurrence and fraction of study area as 0.70. An intersection point value of 0.70 (greater than 0.50) indicates the possible occurrence of high amount of Au mineral within a small area of 0.3 within the Abansuoso Area. This further stipulates that, the P-A plot predicts 0.3 (30 %) of the Abansuoso area as prospective based on BWM-TOPSIS approach, within which 0.7 (70 %) of existing gold occurrences can be found.

Performance assessment of the hybrid BWM-TOPSIS based MPM (shown in Fig. 6) using the ROC curve (see Fig. 9) produced a value of 0.81 for the AUC score. The aforesaid AUC score of 0.81 gives an indication that the predictive performance of the MPM

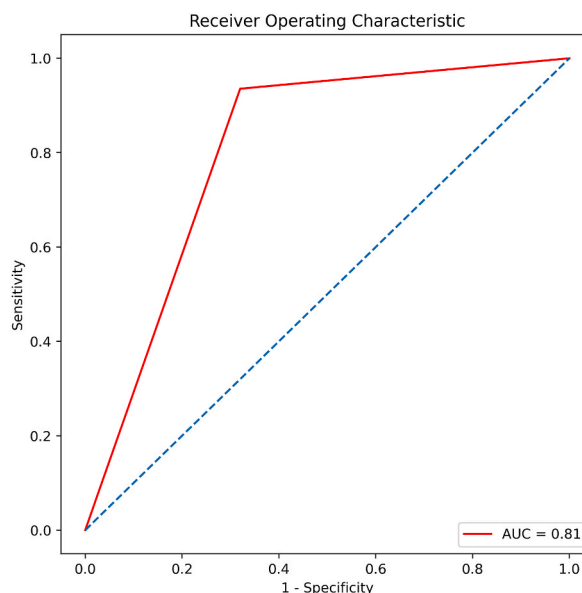


Fig. 9. ROC curve for the MPM produced based on the BWM-TOPSIS procedure.

developed is high (since scores greater than 0.7 depicts models with good accuracy). In view of this, the MPM produced for the study area is efficient and hence the generated MPM's accuracy is very good in relation to the delineated zones of mineral occurrences in the Abansuoso Area.

5. Conclusion

This study's overarching objective was to employ a hybrid BWM-TOPSIS approach to synthesise multi-influencing factors sourced from geoscientific datasets comprising geological and geophysical data to develop a predictive model over the Abansuoso Area of the Ahafoano North District of Ghana's Ashanti Region that characterises prospective zones of mineral occurrences. The best-worst method relies on expert judgment in determining weights for various mineral influencing factors to be integrated towards the production of MPM. In the case of the TOPSIS method, its employability seeks to prioritise and rank the various alternatives within the selected influencing factors that are to be synthesised to generate an MPM. In spite of the capacity of each of these aforementioned methods (BWM and TOPSIS) in combining various influencing factors in generating a prospectivity model, a hybrid of these two methods in carrying out the aforesaid objective produces optimised results by delimiting the target areas to be explored. Furthermore, the use of methods (such as BWM) that integrates factors to generate a predictive model by determining their respective weights based on expert judgments delineate potential zones with relatively wider areal extents. Also, the alternatives (or pixel) ranking methods such as TOPSIS, ranks various pixels devoid of expert judgments on the importance of the factors, whose alternatives are to be ranked. In view of this, the weight of nine mineral influencing factors (geology, distance to faults, lineament density, analytic signal, reduction-to-equator, potassium concentration, uranium concentration, potassium-thorium ratio and uranium-thorium ratio factors) have been determined using the BWM whereas the influence of their alternatives have been computed using the TOPSIS approach to delineate prospective zones of gold mineralisation occurrence in the Abansuoso Area. By implementing the P-A plot technique on the BWM-TOPSIS based MPM generated using the 99 existing locations of Au occurrences, 30 % of the study area was deemed prospective and 70 % of the known Au occurrence were found within the prospective regions. To further analyse the efficacy of the MPM produced, the area under the ROC curve was used. An AUC score of 0.81 was produced, which gives an indication that the MPM developed has a very good accuracy. Given this, the hybrid BWM-TOPSIS approach for mineral prospectivity mapping could be employed as an efficient approach in other studies to guide exploration geoscientists to develop strategies essential for delimiting prospective zones of occurrence of mineral deposits such as gold (as done in this study).

Declarations

We confirm that the manuscript has been read and approved by all the named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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Data availability statement

Data used in this study is available upon reasonable request. Request for data would further require consent from the owners (Ghana Geological Survey Authority and Pelangio Exploration) as authors do not own the datasets used.

CRediT authorship contribution statement

Eric Dominic Forson: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daniel Kwayisi:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Raymond Webrah Kazapoe:** Writing – review & editing. **Clement Ntori:** Writing – review & editing, Methodology. **Solomon Kwasi Adjei:** Writing – review & editing, Methodology. **Abu Mahamuda:** Writing – review & editing. **Shaibu Suleman:** Writing – review & editing, Methodology. **Paul Ameyoe:** Writing – review & editing. **Kennedy Yaw Amedzro:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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